

Refineries are industrial facilities where crude oil is processed and transformed into useful products such as petrol, diesel, kerosene, lubricants, and chemicals. Crude oil, in its natural form, contains a mixture of hydrocarbons and impurities that cannot be used directly. Refineries use advanced chemical and mechanical processes to separate and purify these components. The refining process usually begins with distillation, where crude oil is heated and separated into different fractions based on boiling points. Further treatment processes such as cracking, reforming, and blending improve the quality and usability of the products.

Oil refineries play a major role in supporting transportation, manufacturing, and energy production worldwide. They provide fuels for vehicles, aircraft, ships, and industries, making them essential to modern economies. Many refineries also produce raw materials for plastics, fertilizers, detergents, and synthetic fibers used in everyday life.

However, refineries can have environmental impacts because they release greenhouse gases and other pollutants. To reduce these effects, modern refineries are adopting cleaner technologies, energy-efficient systems, and stricter environmental regulations. As the world moves toward renewable energy, refineries are also exploring sustainable fuel alternatives and greener production methods.